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BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 **MLOPs Report**

Group ID: R26-DS-004

Project Title:

1. IT22118936: Transaction Semantic Reasoning
2. IT22064486: Optimization and Explainable Engine
3. IT22896186: Intelligent Tax Advisory Language Model
4. IT22238580: Personalized Recommendation and Predictive Impact Modelling Engine

		IT22118936	IT22064486	IT22896186	IT22238580
Data Pipeline	**Data Sources:**	The component uses bank statement records containing transaction descriptions, amounts, dates, credit/debit indicators, and bank-specific information. These records are used to identify the likely income category of each transaction. The research also	The component uses Sri Lankan Inland Revenue legislation, including the Inland Revenue Act No. 24 of 2017 and relevant amendments, as its primary data source. Official legal documents and related tax publications provide the evidence for extracting tax rates, relief limits, and eligibility conditions. Synthetic taxpayer	The component uses a versioned corpus of Inland Revenue Department legal documents, including tax legislation and amendments, to answer tax-related questions. It also uses labelled queries for intent classification and evaluation, together	The dataset includes taxpayer financial profiles reflecting realistic financial patterns, structured tax-strategy information, applicable tax rules, and survey responses collected through the Tax-Saving Strategy Adoption Survey. The profiles

		uses labelled transaction records for model training and evaluation, alongside Sri Lankan income-tax legislation and a versioned tax-rule book for determining tax treatment.	profiles and representative tax calculation cases are also used to evaluate the accuracy and traceability of the extracted rules and calculations.	with legal passages retrieved from the document corpus. English and Sinhala-English queries are processed to support user questions, while authoritative legal sources provide evidence for generating and validating tax explanations.	incorporate financial and planning characteristics, such as income levels, deductions, and financial circumstances, to identify eligible strategies and generate personalised recommendations. These profiles, together with rule-derived labels, support model development and evaluation, while the survey provides additional insights into respondents' financial situations and reported tax-strategy adoption considerations.

	Data Preprocessing:	Bank statement data is cleaned and normalized to handle variations in transaction descriptions and bank-specific formats. Relevant text and contextual information are prepared for classification, and transaction records are assigned income-category labels for supervised learning. The dataset is divided into training, validation, and testing subsets, with the recorded experiment using 2,789 training records, 598	Legal documents are processed to extract and organize relevant legislative text into searchable passages. The component uses semantic embeddings and keyword-based retrieval to identify relevant legal evidence. A language model extracts structured tax-rule information from the retrieved passages, after which quotations, source authority, and extracted values are checked before rules are approved for use. Verified rules are organized into assessment-year-specific structures for deterministic tax calculations.	User queries are normalized to handle English and Sinhala-English wording before intent classification and legal retrieval. Relevant legal passages are retrieved from the versioned corpus and reordered using legal-priority information to improve the relevance of the evidence. The selected passages are supplied to a constrained generation process, and symbolic validation checks are applied to identify unsupported or	Taxpayer profiles are transformed into structured financial features, including taxable income, deductions, effective tax rates, and strategy eligibility attributes. The deterministic rules engine filters out strategies that are not applicable before the machine-learning models rank the remaining options. TF-IDF and cosine similarity are used to retrieve relevant strategies, while synthetic profiles are divided into evaluation subsets. Financial assumptions such as

		validation records, and 598 test records.		inconsistent responses. Evidence and validation events are recorded in a Proof Map to support traceability.	salary growth, inflation, and investment returns are varied during Monte Carlo simulation to estimate a range of possible future outcomes.
	Data storage and versioning	Processed transaction records, extracted features, and model artefacts are maintained in structured project storage. PostgreSQL is used for application data persistence, while trained model files and preprocessing configurations are retained for reuse during inference. Maintaining versioned datasets,	The component uses PostgreSQL as its primary database option for document metadata, extracted tax rules, and compiled year-specific rules. Legal source information and verified rule records are maintained so that calculation results can be traced back to their supporting evidence. Database schema changes are managed through Alembic, and the component	The component maintains a versioned legal-document corpus and associated retrieval artefacts to support assessment-year-aware tax guidance. Retrieved passages, source references, and validation events are recorded to preserve the evidence behind generated responses. The	The component stores its datasets, structured taxpayer profiles, trained model artefacts, preprocessing configurations, and evaluation outputs as reusable research files. Ranking and adoption models are maintained separately from the deterministic tax rules and simulation logic. This separation supports

		model bundles, and configuration files supports reproducibility and makes it easier to investigate classification errors.	separates legal document processing, rule extraction, verification, and calculation stages to support maintenance and controlled updates.	Proof Map helps trace the relationship between a user query, retrieved legal evidence, and the resulting advisory response, supporting review when legal sources or validation logic are updated.	repeatable evaluation and helps ensure that changes to recommendation models do not silently alter the tax calculations performed by the rules engine.
Model Development	**Model Selection**	The classification approach combines a multilingual DistilBERT model to predict the likely income category of a bank transaction based on its description and contextual information. A TF-IDF and Logistic Regression model is also available as a	Tax liabilities are calculated through a deterministic tax-rule engine rather than a machine-learning model. Semantic embeddings generated using text-embedding-3-small support legal-text retrieval, while keyword-based retrieval complements semantic search. GPT-4o extracts structured tax-rule	The advisory framework integrates centroid-based intent classification, legal document retrieval, legal-priority reranking, constrained language generation, and symbolic validation. TF-IDF retrieval helps	For personalized strategy ranking, the framework applies LightGBM LambdaMART to prioritise eligible tax-saving strategies according to taxpayer features. TF-IDF and cosine similarity identify textually relevant strategies, and both approaches are

		simpler fallback for classification. This approach supports multilingual transaction descriptions while keeping transaction classification separate from the versioned tax rules that determine the applicable tax treatment.	information from legal passages, but the extracted rules must pass quotation verification and source-review checks before they can be used in calculations.	identify relevant legal evidence, while optional Neo4j enrichment supports the representation of legal relationships. The Think Twice validation mechanism examines responses for potential inconsistencies and unsupported legal assertions before they are presented to users.	combined through a weighted hybrid recommender. In addition, a gradient-boosted classifier estimates adoption likelihood, Monte Carlo simulation models uncertain multi-year financial outcomes, and SHAP explains the factors influencing recommendation rankings.
	Model Training	The multilingual DistilBERT classifier is trained on labelled transaction records prepared through the preprocessing pipeline. Separate training, validation,	Rather than training a predictive model to calculate tax liabilities, the research applies a controlled legal-rule extraction and verification workflow. GPT-4o extracts structured information	The intent classifier is evaluated using labelled tax-related queries, while the retrieval and reranking stages are assessed using query-relevance measures. Retrieved	The LambdaMART ranking model and gradient-boosted adoption classifier are developed using structured taxpayer features and synthetic data. The ranking model

		and test datasets support model development and performance evaluation. The TF-IDF and Logistic Regression approach provides a simpler baseline or fallback for comparison. The recorded experiment evaluates four income categories, so further training and testing are required to establish performance across the complete 20-label taxonomy.	from retrieved legislative passages, and the extracted rules undergo quotation verification, source-authority checks, and review before promotion. The deterministic calculation engine then applies the approved rules for the relevant assessment year. Evaluation includes representative tax cases, relief-limit checks, synthetic taxpayer profiles, and source-traceability assessments.	legal evidence is combined with constrained response generation and symbolic validation. The thesis reports results from a 55-query retrieval evaluation and a separate 40-case adversarial validation suite. These results should be interpreted cautiously because the datasets are limited and some raw evaluation records remain unavailable.	learns to prioritize eligible strategies, while the adoption classifier relies on a rule-derived proxy label because validated real-world adoption records are unavailable. The Monte Carlo engine compares baseline and strategy-adoption scenarios under varying financial assumptions, and SHAP identifies the features influencing recommendation rankings.
Model Integration	**Tools used**	Python supports backend processing and model integration, while	The technology stack includes Python 3.11, FastAPI, SQLAlchemy, Alembic, and	Python and FastAPI support the advisory backend, including query	Python provides the foundation for recommendation ranking, adoption

		FastAPI provides API endpoints and PostgreSQL supports data persistence. The implementation combines multilingual DistilBERT, TF-IDF, Logistic Regression, and supporting machine-learning libraries. React provides the auditor and taxpayer interfaces, allowing transaction classifications and review information to be incorporated into the wider TAX Advisory workflow.	PostgreSQL for backend processing and database management. OpenAI's text-embedding-3-small supports semantic retrieval, while GPT-4o extracts structured tax-rule information from legal passages. React 19, TypeScript, Vite, and Tailwind CSS support the frontend interface for presenting tax calculations and their supporting legal evidence.	processing, intent classification, legal retrieval, response generation, and validation. TF-IDF supports text retrieval, while language-model services generate responses grounded in retrieved legal evidence. Optional Neo4j integration supports legal relationship enrichment, and the Proof Map records supporting evidence and validation events for inspection.	prediction, financial simulation, and explainability. LightGBM supports LambdaMART ranking, while TF-IDF and cosine similarity provide text-based strategy retrieval. Additional tools support gradient-boosted classification, Monte Carlo simulation, and SHAP explanations. REST APIs connect the backend workflow to a React dashboard containing taxpayer and auditor views.
Model Deployment	**Testing Environments**	The classification pipeline is tested in a local development environment using	Representative Sri Lankan tax calculation cases, synthetic taxpayer profiles, and	Labelled tax-related queries, retrieval-quality measurements, and	Held-out taxpayer profiles, ranking and classification metrics, and

		separate training, validation, and test datasets. Performance is evaluated on held-out transaction records, with the recorded experiment containing 598 test records. Backend and user-interface checks also examine whether predicted categories, tax-rule processing, and review controls work together correctly. Broader validation remains necessary because the recorded experiment does not cover the full 20-label taxonomy.	legal-rule verification scenarios are used to assess the implementation. Testing checks whether extracted tax rates and relief limits correspond to supporting legislative sources and whether calculations produce expected results for the relevant assessment year. Retrieval and source-traceability checks examine whether explanations can be linked to appropriate legal passages. These evaluations do not establish expert-level accuracy for every taxpayer situation.	adversarial validation cases are used to examine the advisory workflow. Testing evaluates whether relevant legal passages are retrieved, whether legal-priority reranking improves retrieval quality, and whether symbolic validation identifies unsupported responses. The reported evaluations include 55 labelled queries and a separate 40-case adversarial suite. Additional independent testing is needed because the datasets are limited and some	simulation-based tests support evaluation of the recommendation pipeline. Different recommendation methods are compared to assess the contribution of ranking, retrieval, and hybrid recommendation. Monte Carlo simulation is examined through financial plausibility and convergence checks, while SHAP explanations help identify the features influencing recommendations. Since the adoption model relies on proxy labels, its

				raw annotation records remain unavailable.	performance does not establish how real taxpayers will behave.
	Deployment Platform	A FastAPI backend, PostgreSQL persistence layer, and React-based interfaces form the main implementation environment. The trained multilingual classification model is integrated into the transaction-processing workflow so that bank statement records can be categorised and reviewed. The thesis describes a research implementation, but it does not establish deployment as a	The implementation consists of a Python and FastAPI backend connected to PostgreSQL, together with a React-based frontend. API endpoints provide access to document processing, legal retrieval, rule verification, and deterministic tax calculation. The environment supports local development and testing; production deployment as a public service is not established in the thesis.	The advisory service connects query processing, legal retrieval, reranking, constrained generation, and symbolic validation through a backend implementation. FastAPI provides the service interface, while retrieved legal evidence and validation information support traceable responses. The component is designed for integration with the wider TAX	A Python backend, REST APIs, and a React dashboard provide the main environment for the recommendation workflow. The backend processes taxpayer profiles, filters eligible strategies, generates rankings, estimates adoption likelihood, and calculates simulated financial outcomes. The dashboard presents recommendations, explanations, and uncertainty ranges for taxpayer and auditor review. The

		fully operational production tax service.		Advisory application, but the thesis does not establish deployment as a fully validated public-facing tax advisory service.	implementation remains a research component rather than a fully validated production tax platform.
	Deployment Method	The backend loads the trained classification model and its preprocessing configuration when transaction classification is required. It processes transaction information, generates a predicted income-category label, and passes the result into the wider workflow for rule-based tax	API endpoints expose the functionality required for legal document processing, evidence retrieval, rule verification, and deterministic tax calculation. Verified tax rules are compiled into assessment-year-specific structures before being applied by the calculation engine. Results are presented with calculation traces and supporting legal passages so users can inspect the basis of a	The advisory service normalizes a user's question, identifies its intent, retrieves relevant legal passages, and reorders the evidence before generating a response. Symbolic validation checks help reduce unsupported legal claims and inconsistencies. The Proof Map preserves the connection between	The service receives an established taxpayer financial profile and applies deterministic eligibility filtering before invoking the recommendation models. The hybrid recommender ranks eligible strategies, the adoption model estimates their likelihood of adoption, and the Monte Carlo engine projects possible

		treatment and review. Uncertain cases should be reviewed rather than automatically released as confirmed taxpayer information. This approach separates model predictions from legally significant tax decisions.	tax result. Language-model extraction remains separate from the final tax calculation.	retrieved evidence and the generated response, making the output easier to inspect. Questions that cannot be answered reliably from available evidence require clarification or further review.	financial outcomes. SHAP explanations and uncertainty ranges are then made available through the dashboard. The rules engine remains responsible for tax calculations, while machine knowledge supports personalization.
Future Enhancements	**Model Improvement:**	Further development should expand the transaction dataset to cover more banks, transaction descriptions, income categories, and financial situations. The classifier should be evaluated against the full 20-label taxonomy	Future work should broaden legal-document coverage and incorporate legislative amendments to ensure that tax rules remain accurate as Sri Lankan tax legislation changes. More extensive testing should cover assessment years, relief limits, income types, and complex taxpayer	Future improvements should include larger and more representative taxpayer datasets to reduce reliance on synthetic profiles and rule-derived adoption labels. The adoption-likelihood model should be evaluated	Future work should expand the legal corpus to cover additional legislative amendments and improve support for different assessment years and English or Sinhala-English queries. Larger, independently reviewed query

		rather than only the current four-class test setting. Additional human-reviewed examples, stronger handling of unfamiliar transaction descriptions, and better confidence-based review mechanisms could improve reliability. Further testing should examine classification performance across different banks and transaction types before wider deployment.	situations. Independent evaluation by qualified tax professionals would help establish whether extracted rules and calculation explanations are legally accurate. Improvements to quotation verification, source traceability, and automated detection of outdated rules could strengthen reliability and maintainability.	against observed behaviour when suitable data becomes available. Recommendation quality could be improved through broader strategy coverage, further comparison of hybrid ranking methods, and stronger validation of SHAP explanations. The Monte Carlo engine should also be assessed under a wider range of realistic financial assumptions to improve the usefulness of projected outcomes.	datasets would help evaluate intent classification, retrieval quality, and the reliability of generated answers. Further improvements should strengthen symbolic validation, reduce unsupported legal assertions, and improve the handling of ambiguous questions. Regular source updates and independent legal review would help maintain the accuracy, traceability, and reliability of the advisory service.